

Java Arrays, Strings & Collections — Quick Reference

1D/2D Arrays, java.util.Arrays, ArrayList, LinkedList, String methods

#	Topic	Import Syntax	Method / Syntax	Usage	Example Input	Example Output
1	1D Array	No import	arr.length	Find size	{10, 20, 30}	3
2	1D Array	No import	arr[i]	Access element	arr[1]	20
3	1D Array	No import	arr[i] = x	Update element	arr[1]=50	{10, 50, 30}
4	1D Array	No import	for(int x : arr)	Traverse	{1, 2, 3}	1 2 3
5	2D Array	No import	arr.length	Number of rows	{{1, 2}, {3, 4}}	2
6	2D Array	No import	arr[i].length	Number of columns	{{1, 2}, {3, 4}}	2
7	2D Array	No import	arr[i][j]	Access element	arr[1][0]	3
8	2D Array	No import	Nested for	Traverse matrix	{{1, 2}, {3, 4}}	1 2 3 4
9	Arrays	import java.util.Arrays;	Arrays.sort(arr)	Sort array	{5, 2, 1}	{1, 2, 5}
10	Arrays	import java.util.Arrays;	Arrays.toString(arr)	Print 1D array	{1, 2, 3}	[1, 2, 3]
11	Arrays	import java.util.Arrays;	Arrays.deepToString(arr)	Print 2D array	{{1, 2}, {3, 4}}	[[1, 2], [3, 4]]
12	Arrays	import java.util.Arrays;	Arrays.fill(arr, x)	Fill array	{1, 2, 3}, 0	{0, 0, 0}
13	Arrays	import java.util.Arrays;	Arrays.copyOf(arr, n)	Copy array	{1, 2, 3}, 5	{1, 2, 3, 0, 0}
14	Arrays	import java.util.Arrays;	Arrays.copyOfRange(arr, l, r)	Copy range	{1, 2, 3, 4}, 1, 3	{2, 3}
15	Arrays	import java.util.Arrays;	Arrays.equals(a, b)	Compare arrays	{1, 2}, {1, 2}	true
16	ArrayList	import java.util.ArrayList;	ArrayList list = new ArrayList<>();	Create list	—	[]
17	ArrayList	import java.util.ArrayList;	list.add(x)	Add element	10	[10]
18	ArrayList	import java.util.ArrayList;	list.add(i, x)	Add at index	1, 20	[10, 20]
19	ArrayList	import java.util.ArrayList;	list.get(i)	Get element	list.get(1)	20
20	ArrayList	import java.util.ArrayList;	list.set(i, x)	Update element	set(1, 50)	[10, 50]
21	ArrayList	import java.util.ArrayList;	list.remove(i)	Remove by index	remove(1)	[10, 30]
22	ArrayList	import java.util.ArrayList;	list.remove(Integer.valueOf(x))	Remove by value	remove(20)	[10, 30]
23	ArrayList	import java.util.ArrayList;	list.size()	Get size	[10, 20, 30]	3
24	ArrayList	import java.util.ArrayList;	list.contains(x)	Check element	contains(20)	true

#	Topic	Import Syntax	Method / Syntax	Usage	Example Input	Example Output
25	ArrayList	import java.util.ArrayList;	list.clear()	Remove all	[1, 2, 3]	[]
26	ArrayList	import java.util.ArrayList;	list.isEmpty()	Check empty	[]	true
27	LinkedList	import java.util.LinkedList;	LinkedList list = new LinkedList<>();	Create list	–	[]
28	LinkedList	import java.util.LinkedList;	list.add(x)	Add element	10	[10]
29	LinkedList	import java.util.LinkedList;	list.addFirst(x)	Add beginning	5	[5, 10]
30	LinkedList	import java.util.LinkedList;	list.addLast(x)	Add end	20	[5, 10, 20]
31	LinkedList	import java.util.LinkedList;	list.get(i)	Get element	get(1)	10
32	LinkedList	import java.util.LinkedList;	list.getFirst()	Get first	–	5
33	LinkedList	import java.util.LinkedList;	list.getLast()	Get last	–	20
34	LinkedList	import java.util.LinkedList;	list.remove(i)	Remove index	remove(1)	[5, 20]
35	LinkedList	import java.util.LinkedList;	list.removeFirst()	Remove first	–	[10, 20]
36	LinkedList	import java.util.LinkedList;	list.removeLast()	Remove last	–	[5, 10]
37	LinkedList	import java.util.LinkedList;	list.size()	Get size	[5, 10, 20]	3
38	LinkedList	import java.util.LinkedList;	list.contains(x)	Check element	contains(10)	true
39	String	No import	s.length()	Find length	"Hello"	5
40	String	No import	s.charAt(i)	Get character	charAt(1)	e
41	String	No import	s.substring(l, r)	Extract substring	"Hello", 0, 3	Hel
42	String	No import	s.equals(t)	Compare strings	"abc", "abc"	true
43	String	No import	s.equalsIgnoreCase(t)	Compare ignoring case	"ABC", "abc"	true
44	String	No import	s.contains(x)	Check substring	"hello", "ell"	true
45	String	No import	s.indexOf(x)	First occurrence	"hello", 'l'	2
46	String	No import	s.lastIndexOf(x)	Last occurrence	"hello", 'l'	3
47	String	No import	s.toCharArray()	String → char array	"abc"	[a, b, c]
48	String	No import	s.toLowerCase()	Convert lowercase	"HELLO"	"hello"

#	Topic	Import Syntax	Method / Syntax	Usage	Example Input	Example Output
49	String	No import	s.toUpperCase()	Convert uppercase	"hello"	"HELLO"
50	String	No import	s.trim()	Remove outer spaces	" hi "	"hi"
51	String	No import	s.split(" ")	Split string	"a b c"	["a", "b", "c"]